Complex Manifolds

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Abstract

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1 Feb 17

Let $\mathrm{Gr}(k,n)$ denote the Grassmannian of k-planes in V, a real or complex vector space of dimension n. We will not distinguish since the statements we make will be true for both cases. For the real Grassmannian, we take homology with $\mathbb{Z}/2$ coefficients, and for the complex Grassmannian, we take homology with $\mathbb{Z}$ coefficients.

Theorem 1.1. For a CW complex X, $\mathbb{R}, \mathbb{C}, \mathbb{H}$-vector bundles over X up to isomorphism are classified by homotopy classes of maps to the infinite Grassmannian $Gr_k(\mathbb{C}^\infty) = \lim_{n\to\infty} Gr_k(\mathbb{C}^n)$.

Remark 1.2. $\mathrm{Sp}(n) = U(2n) \cap GL(n,\mathbb{H})$ is the compact symplectic group acts on $\mathbb{H}^n$ by left multiplication and commutes with the right $\mathbb{H}$-action which is the structure map for $\mathbb{H}$-vector bundles.

Recall that the Schubert stratification is given by row echelon form for matrices. Let $T \subset V$ be a subspace of dimension k. Write T as the row space of a $k \times n$ matrix. We can perform row operations to put it in row echelon form, and this depends only on T. In particular, T has a canonical basis but it does not depend continuously on T. It may jump whenever the pivots move. The Schubert stratification is given by the locus where the pivots are fixed. Therefore, it is enough to give an increasing sequence σ_p of pivot positions or a sequence s of 0's and 1's of length n with exactly k 1's.

Example 1.3. Let us row reduce from right to left. For example in $\mathrm{Gr}(2,4)$

$$\begin{bmatrix} * & 0 & * & 1 \\ * & 1 & 0 & 0 \end{bmatrix}$$

corresponds to the sequence $s = (0,1,0,1)$ and the pivot positions $\sigma_1 = 2, \sigma_2 = 4$. The dimension is given by

$$\dim S_\sigma = \sum_{p=1}^{k} (\sigma_p - p) = (2-1) + (4-2) = 3 = \sum_{1s\in s} \text{total number of 0s to the left} = 1+2$$

There are two inclusions called i, j of $\mathrm{Gr}(k,n)$ into $\mathrm{Gr}(k,n+1)$ and $\mathrm{Gr}(k+1,n+1)$ given by $T \mapsto T \oplus e_{n+1}$ and $T \mapsto e_0 \oplus T$ respectively.

1.1 Some generalities

These cells define a CW decomposition. They give a basis of $\mathbb{Z}$ homology for $\mathbb{C}, \mathbb{H}$ or $\mathbb{Z}/2$ homology for $\mathbb{R}$. It is obvious for $\mathbb{C}, \mathbb{H}$ because there are no attaching maps, but for $\mathbb{R}$ you have to argue that the attaching maps have even degree. The closure of S_σ is the union of cells S_τ if $\tau_p \leq \sigma_p$ for all p< meaning no pivot of σ is to the left of the corresponding pivot of τ. Poincare duality is implemented by reverseing the sequence of $0s$ and $1s$, reverse the order of the basis and check the intersectino of the cells. Most $\overline{S_\sigma}$ are singular but the singularities have codimension at least 2 and thusd efine fundamental classes and can use the intersection pairing to check poincare duality. In particular they are normal varieties.

1.2 The inclusions i,j

What are the smallest cells i, j are missing? For i we are missing

$$\left[F_k | M | e_0\right]$$

where F_k is the antidiagonal identity, M is stars in the top row and 0s elsewhere, and e_0 is $1, 0, \ldots, 0$ in the last column. This corresponds to the sequence $s = 11\ldots10\ldots01$ and is of dimension $n-k$.

For j we are missing

$$\left[* | F_{k+1} | 0\right]$$

corresponds to the sequence $01\ldots10\ldots0$ and is of dimension $k+1$.

Corollary 1.4 (Approximation theorem)**.** If $\dim X < n-k$ or $\dim X < 2(n-k)$ in the complex case, then a map $X \to \mathrm{Gr}_{\mathbb{R}}(k, n+1)$ or $X \to \mathrm{Gr}_{\mathbb{C}}(k, n+1)$ can be homotoped to a map $X \to \mathrm{Gr}(k, n)$. In particular $\mathrm{Gr}(k, n+1)/\mathrm{Gr}(k, n)$ is $(n-k-1)$-connected or $(2(n-k)-1)$-connected in the complex case.

If $\dim X < k+1$ or $\dim X < 2(k+1)$ in the complex case, then a map $X \to \mathrm{Gr}_{\mathbb{R}}(k+1, n+1)$ or $X \to \mathrm{Gr}_{\mathbb{C}}(k+1, n+1)$ can be homotoped to a map $X \to \mathrm{Gr}(k, n)$. In particular $\mathrm{Gr}(k+1, n+1)/\mathrm{Gr}(k, n)$ is k-connected or $2k+1$-connected in the complex case.

Corollary 1.5 (Stability for classification of vector bundles)**.** If $\dim X < n-k-1$ or $\dim X < 2(n-k)-1$ in the complex case, then $\mathrm{Vect}^k(X) = [X, \mathrm{Gr}(k, n)]$.

Note that we lost one dimension because we need to leave room for the homotopy.

Corollary 1.6. If $\dim X < k+1$ or $\dim X < 2(k+1)$ in the complex case, then any vector bundle of rank $k+1$ has a nonvanishing section.

This also follows from transversality.

1.3 Stiefel manifodls and classifing spaces

Recall that $T^k \to \mathrm{Gr}(k,n)$ is the tautological subbundle of $\mathbb{R}^n$ or $\mathbb{C}^n$. We have the frame bundle $\mathrm{St}(k,n)$ given by basis of k-planes. Gram Schmidt tells us that orthonormal frames $\hookrightarrow$ frames is a homotopy equivalence so we have $\mathrm{St}(k,n) \to \mathrm{Gr}(k,n)$ is a principal GL_k-bundle and $\mathrm{St}^{on}(k,n) \to \mathrm{Gr}(k,n)$ is a principal $O(k)$-bundle or $U(k)$-bundle.

What is the codimension of the complement of $\mathrm{St}(k,n)$ in $\mathrm{Mat}_{k,n}$? Well $\mathrm{Mat}(k,n)$ is stratified by rank and the largest stratum in the complement of $\mathrm{St}(k,n)$ is given by rank $k-1$ matrices which has codimension $n-k+1$.

Corollary 1.7. If $\dim X < n-k$ or $\dim X < 2(n-k)+1$ then $[X, \mathrm{St}(k,n)] = *$. In particiular $\mathrm{St}(k,\infty)$ and $\mathrm{St}^{on}(k,\infty)$ are contractible.

Thus we have stumbled upon the fact that $\mathrm{St}^{on}(k,\infty)$ is a contractible space with a free $O(k)$ or $U(k)$ action and $\mathrm{Gr}(k,\infty)$ is the quotient.

Definition 1.8. Let G be a topological group which is nice in the sense that it has the homotopy type of a CW complex. A CW complex with a principal G bundle $EG \to BG$ with contractible total space is called a **classifing space** for G.

Theorem 1.9. $EG \to BG$ is unique up to homology equivalence, canonical up to homotopy if you choose the base point $*$ in BG and identify the fiber over $*$ with G. Moreover, isomorphism classes of principal G-bundles over a CW complex $X, *$ with trivialization over $*$ are in bijection with homotopy classes of pointed maps $[X, BG]$.

In fact (2) implies (1) because we have identified BG as a functor out of the homotopy category of pointed CW complexes, and the functor is unique up to natural isomorphism.

Remark 1.10. G is connected if and only if BG is simply connected. Extreme cases are G is discrete, then EG is the universal cover of BG.

Remark 1.11. G discrete, X is connected. Then $[X, BG]$ is the set of homomorphisms $\pi_1(X) \to G$ up to conjugation, whereas $[X, BG]_* = \mathrm{Hom}(\pi_1(X), G)$. The first is isomorphism classes of principal G-bundles, the second is isomorphism classes of principal G-bundles with a trivialization at the base point.

Next we are have two models of $BU(k) \to BU(k+1)$. One is the obvious inclusion, and the other model gives it as a fiber bundle with fiber S^{2k+1}.

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Last time we showed that $\mathrm{Gr}_C(n,\infty)$ is a model for $BU(n)$. Note that there are parallel stories for $O(n)$ over $\mathbb{R}$ and $\mathrm{Sp}(n)$ over $\mathbb{H}$ but we will now live over $\mathbb{C}$.

The Grassmannian $\mathrm{Gr}(n,\infty)$ classifies iso classes of rank n complex vector bundles, and we have a principal $U(n)$-bundle $\mathrm{St}^{on}(n,\infty) \to \mathrm{Gr}(n,\infty)$ which is the universal principal $U(n)$-bundle with total space contractible.

We also know that we have embeddings $\mathrm{Gr}(n,\infty) \to \mathrm{Gr}(n+1,\infty)$ which is highly connected. One can think of inductively building $BU(n+1)$ from $BU(n)$ by attaching an extra cell of real dim $2n+2$ ($n+1$, $4n+4$ in the real and quaternionic cases) at each step.

Example 2.1. If $n = 0$ then $\mathrm{Gr}(1,\infty) = k\mathbb{P}^\infty$. $\mathbb{HP}^\infty$ is $BSU(2)$ and the homotopy equivalence $SU(2) \hookrightarrow \mathrm{GL}_1(\mathbb{H})$ shows that $\mathrm{SU}(2)$ bundles are the same as $\mathbb{H}$-line bundles.

Observe that $U(n) \hookrightarrow U(n+1)$ has quotient S^{2n+1} from the action of $U(n+1)$ on the unit sphere in $\mathbb{C}^{n+1}$. This gives a fiber bundle $S^{2n+1} \to BU(n) \to BU(n+1)$ which is the other model of the inclusion $BU(n) \to BU(n+1)$.

Proposition 2.2. The total space of the fiber bundle $\mathrm{St}^{on}(n+1,\infty) \times S^{2n+1} \to \mathrm{Gr}(n+1,\infty)$ is homotopy equivalent to $\mathrm{Gr}(n,\infty)$ and the fiber bundle map is homotopic to the inclusion $\mathrm{Gr}(n,\infty) \to \mathrm{Gr}(n+1,\infty)$. Note that there is a natural lift $\mathrm{Gr}(n,\infty) \to \mathrm{St}^{on}(n+1,\infty)$ which is a homotopy equivalence.

This gives a nonprincipal fiber bundle $S^{2n+1} \to \mathrm{Gr}(n,\infty) \to \mathrm{Gr}(n+1,\infty)$.

Proof. A map $X \to \mathrm{St}_{\mathbb{C}}(n+1,\infty) \times_{U(n+1)} S^{2n+1}$ classifies up to isomorphism a complex vector bundle of rank $n+1$ together with a section of unit norm. This is equivalent (up to isomorphism and rescaling of hermitian metrics (which is a contractible space)) to $E^{n+1} \to X, s$ where s is a nonvanishing section. This is equivalent to a rank n vector bundle by taking the quotient. Thus the target space $\mathrm{St}_{\mathbb{C}}(n+1,\infty) \times_{U(n+1)} S^{2n+1}$ is a classifying space for rank n vector bundles, and thus homotopy equivalent to $\mathrm{Gr}(n,\infty)$. The map to the base $\mathrm{Gr}(n+1,\infty)$ classifies the stabilized bundle of rank $n+1$ and so is homotopy equivalent to natural map. $\square$

Now let us consider the general construction, note that $S^{2n+1} = U(n+1)/U(n)$ is a homogeneous space. Let G be a nice topological group. Let H be a closed subgroup of G.

Let $EG \to BG$ be the classifiing space for principal G-bundles. Suppose that $H \to G \to G/H$ is a prinicpal H-bundle, in particular locally trivial. This is always true for Lie groups.

Theorem 2.3. EG/H is a classifing space for H. In particular, in the diagram below, there is a unique up to homotopy lifting of this map which is a homotopy equivalence.

$$\begin{tikzcd} EG \arrow[r, "f"] \arrow[d, "p"'] & EG/H \arrow[d, "q"] \\ BH \arrow[r] \arrow[ur, dotted, "h"] & BG \end{tikzcd}$$

Proof. First we need a lemma. Isomorphism classes of principal H-bundles on X are the same as isomorphism classes of principal G bundles together with a section of associated G/H-bundle up to homotopy of sections. This is saying that if $P \to X$ is a principal H-bundle, then you can form $Q = P \times_H G$ which is a principal G-bundle. A section of Q/H is the same as an isomorphism of Q with a bundle of the form $P \times_H G$. $\square$

Now we can build classifing spaces for all Lie groups. Recall from the Peter Weyl theorem that every compact Lie group has a faithful finite dimensional unitary representation, faithful being the key word here. This gives us $G \hookrightarrow U(n)$ for some n. So we can build a classifing space for G using our favorite model for $BU(n)$ and then dividing by G

$$EU(n) \to EU(n)/G = BG$$

and the good thing here is that this makes BG into an increasing union of compact manifolds which stabilize in homology. This is the story of classifing spaces for compact Lie groups.

Remark 2.4 (Slice theorem for quotients)**.** When groups act on spaces, you have to be careful when you take quotients. It might not be Hausdorff for example. Let G be a possibly noncompact Lie group acting properly on a normal Hausdorff space X. (Normal means that have seperation of closed sets by open neighborhoods. Urysohn's lemma says that this means we can separate by values of continuous functions.) Then there is a slice through each point: for each $x \in X$, there is a G_x-invariant neighborhood S of x such that the natural map $G \times_{G_x} S \to X$ is an equivariant homeomorphism onto a G-invariant neighborhood of the orbit $G \cdot x$. This gives a linear local model using the normal space at the orbit. insert more remark

2.1 Three classical theorems

Theorem 2.5 (Leray Hirsch)**.** Let $F \to E \xrightarrow{p} B$ be a connected fiber bundle over a CW base. Let R be a coefficient ring for cohomology, i.e a quotient of $\mathbb{Z}$ or somehting between $\mathbb{Z}$ or $\mathbb{Q}$. Suppose $H^*(F;R)$ is a free R-module on $\alpha_1, \ldots, \alpha_k$ and that there exist classes $a_1, \ldots, a_k \in H^*(E;R)$ such that $a_i|_F = \alpha_i$. Note that this is a strong assumption because there is some subtely about identification of F with the fiber over a point in B.

Then the map $H^*(F;R) \otimes_R H^*(B;R) \to H^*(E;R)$ given by $x \otimes y \mapsto p^*y \cup a_i$ is an isomorphism of R-modules.

Remark 2.6. This recovers Kunneth formula for the product $E = F \times B$ by taking $a_i = \alpha_i \otimes 1$. If F has R-torsion, this cannot work because of Kunneth.

Example 2.7 (Nonexamples)**.** $\mathbb{R} \to S^1$. $S^3 \subset \mathbb{C}^2 \to S^2 = \mathbb{CP}^1$ the Hopf fibration. In the second case, we compute

$$H^*(\S^3; \mathbb{Q}) = \mathbb{Q} \oplus \mathbb{Q}[-3]$$
$$H^*(S^2; \mathbb{Q}) \otimes H^*(\S^1; \mathbb{Q}) = \mathbb{Q} \oplus \mathbb{Q}[-1] \oplus \mathbb{Q}[-2] \oplus \mathbb{Q}[-3]$$

What is happening is that there is a grid

$$\begin{bmatrix} \mathbb{Z} & 0 & \mathbb{Z} \\ \mathbb{Z} & 0 & \mathbb{Z} \end{bmatrix}$$

and there is a differential d_2 from the $\mathbb{Z}$ in the bottom right to the $\mathbb{Z}$ in the top left which kills the $\mathbb{Z}$ in the bottom right and the $\mathbb{Z}$ in the top left.

Proof. Idea of proof. Build B from its skeleta B_n and check that the map remains an iso at each step from the LES and the five lemma.

$$H^{*-1}(B_n) \to H^*(B_{n+1}, B_n) \to H^*(B_{n+1}) \to H^*(B_n) \to H^{*+1}(B_{n+1}, B_n)$$

Tensor with $H^*(F)$. Also have a sequence for $\pi^{-1}(B_n)$. Get a map from $H^*(F) \otimes H^*(B_n)$ to $H^*(\pi^{-1}(B_n))$ and check that it is an iso at each step. The key iput is that we have a map between the original complexes. $\square$

Variant with the same proof: $E' \hookrightarrow E$ two fiber bundles over B. Use $H^*(F, F'; R)$ and $H^*(E, E'; R)$. Same hypothesis and conclusions.

Example 2.8. $U(n-1) \to U(n) \to S^{2n-1}$ will work, even though $U(n)$ is not homotopy equivlent to a product of odd spheres. Rationally it is, but integality is the issue.

The second classical theorem is an application of Leray.

Theorem 2.9 (Thom isomorphism theorem)**.** Let $E^r \to X$ be a real vector bundle of rank r with a metric. Let $D(E)$ be the unit disk bundle and $\partial D(E)$ be the unit sphere bundle. **Assume** that E is relatively orientable for a ring R. Then there exist a class $\theta \in H^r(D(E), \partial D(E); R)$ which restricts to the preferred generator in each fiber. Consequently by Leray, θ is such that the map $H^*(X; R) \to H^{*+r}(D(E), \partial D(E); R)$ given by $x \mapsto x \cup \theta$ is an isomorphism.

E is relatively orientable means that we can choose orientations of the fibers which vary continuously over the base. If $R = \mathbb{Z}/2$ there is no condition. If X is an oriented manifold then $(D^r, \partial D^r)$ becomes an oriented manifold with boundary, and then E is Poincare dual to the zero section of E. θ is called the Thom class of E and the isomorphism is called the Thom isomorphism.

The theorem is ok if you just have disk bundles and its boundary. One has disk bundles which do not come from vector bundles. The third theorem is the Gysin sequence but we will not get there today. This will show that $H^*(\mathrm{Gr}(n,\infty))$ is a polynomial ring on n generators of degree $2, 4, \ldots, 2n$ called the Chern classes.

Definition 2.10. The Euler class of E^r is the image of θ under the map $H^r(D(E), \partial D(E); R) \to H^r(D(E); R) = H^r($ given by inclusion.

If X is compact oriented manifold, then $PD(e)$ is the zero locus of a gneric section (transversal intersection with the zero section of E).

2.2 Line bundles and chern classes

Recall isomorpihsm classes of real, complex, quaternionic line bundles are classified by $BO(1) = \mathbb{RP}^\infty$, $BU(1) = \mathbb{CP}^\infty$, $B\,\mathrm{SU}(2) = \mathbb{HP}^\infty$ respectively, and the cohomologies are (use the positive generators)

$$\begin{aligned} H^*(\mathbb{RP}^\infty; \mathbb{Z}/2) &= \lim_{n\to\infty} H^*(\mathbb{RP}^n; \mathbb{Z}/2) = \mathbb{Z}/2[w], \deg w = 1 \\ H^*(\mathbb{CP}^\infty; \mathbb{Z}) &= \lim_{n\to\infty} H^*(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}[c], \deg c = 2 \\ H^*(\mathbb{HP}^\infty; \mathbb{Z}) &= \lim_{n\to\infty} H^*(\mathbb{HP}^n; \mathbb{Z}) = \mathbb{Z}[p], \deg p = 4 \end{aligned}$$

Proposition 2.11. w, c are the Euler classes of the respective line bundles over $\mathbb{RP}^\infty$ and $\mathbb{CP}^\infty$ respectively. For $\mathbb{CP}^\infty$, we use $\mathcal{O}(1)$ the dual of the tautological sub line bundle.

Proof. A linear section has zero locus a hyperplane, the PD of the positive generator of cohomology. $\square$

Definition 2.12 (Characteristic classes)**.** For a real line bundle $L \to X$ classified by $k_L : X \to \mathbb{RP}^\infty$, the Stiefel-Whitney class of L is $w(L) = k_L^* w$. For a complex line bundle $L \to X$ classified by $k_L : X \to \mathbb{CP}^\infty$, the Chern class of L is $c(L) = k_L^* c$.

Theorem 2.13. $w_1(L) = w(L)$ and $c_1(L) = c(L)$ determine the isomorphism class of L. Moreover,

$$\begin{aligned} w_1(L \otimes L') &= w_1(L) + w_1(L') \\ c_1(L \otimes L') &= c_1(L) + c_1(L') \end{aligned}$$

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3.1 Suspensions and smash product

You should think of smash product as a tensor product. This becomes precise in linearized, i.e. stable homotopy theory. Also we need base points to talk about smash products.

Example 3.1. $I, 0 \wedge I, 0$ is not very interesting. $S^m \wedge S^n$ is S^{m+n}. To see this, identify $S^m, *$ with the unit cube $I^m, \partial I^m$ so that

$$\begin{aligned} S^m \wedge S^n &= (I^m \wedge I^n)/(\partial I^m \wedge I^n \cup I^m \wedge \partial I^n) \\ &\cong I^{m+n}/\partial I^{m+n} \cong S^{m+n} \end{aligned}$$

Definition 3.2. The reduced suspension of a pointed space $X, *$ is $\Sigma X = S^1 \wedge X$. If X does not have a base point, we can add one by taking the disjoint union with a point and declaring that to be the base point, the space X_+. Then declare the unreduced suspension ΣX to be the reduced suspension of X_+.

To think of a suspension, you can think of a double cone on X and collapse a copy of X that stretches from the top of the cone to the bottom.

Theorem 3.3. X based space. Then G-bundles on ΣX with a trivialization at the based point are in bijection with based homotopy classes of maps $[X, G]_*$.

Proof. Bundles are trivial on the cones over X. Only info is the gluing along $X, *$ which is given by a map $X \to G$ up to homotopy. $\square$

Corollary 3.4. $[\mathrm{Bun}_G(\Sigma X)]_* = [\Sigma X, BG]_* = [X, \Sigma BG]_*$

Recall that $\mathrm{Map}_*(\Sigma X, Y) \cong \mathrm{Map}_*(X, \Omega Y)$ where ΩY is the based loop space of Y if X is Hausdorff.

Corollary 3.5. The classifing map of the standard G bundle $\Sigma G \to BG$ induces a homotopy equivalence $G \to \Omega BG$ which preserves the group structure up to homotopy.

3.2 Line bundles on spheres

For $n > 0$ then we can compute

$$[\mathrm{Bun}_{\mathrm{GL}(1)}(S^n)]_* = [S^{n-1}, \mathrm{GL}(1)]_*$$

Note that $[S^{n-1}, \pm 1] = \pm 1$ if $n = 1$ and $*$ if $n > 1$. Also note that $[S^{n-1}, S^1] = \mathbb{Z}$ if $n = 2$ and $*$ if $n \neq 2$. This is because any S^{n-1} with $n > 2$ is simply connected so maps lift to $\mathbb{R}$, the universal cover of S^1, which is contractible.

Remark 3.6. The Eilenberg-MacLane space of type π, n where π is an abelian group and n is an integer is the unique up to homotopy, connected based space $K(\pi, n)$ such that $\pi_n(K(\pi, n)) = \pi$ and $\pi_m(K(\pi, n)) = 0$ for $m \neq n$.

We have met a couple examples of Eilenberg-MacLane spaces: $K(\mathbb{Z}/2, 1) = \mathbb{RP}^\infty$ and $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$. S^1 is a $K(\mathbb{Z}, 1)$. Note that $\mathbb{HP}^\infty$ is not a $K(\mathbb{Z}, 4)$ because there are higher homotopy groups. A cool example is that $K(\mathbb{Z}/n, 1)$ is the total space of the sphere bundle of $\mathcal{O}(1)^n$ over $\mathbb{CP}^\infty$, also known as $S^\infty/\mathbb{Z}/n$ where $\mathbb{Z}/n$ acts by n-th roots of unity on the unit sphere in $\mathbb{C}^\infty$.

Theorem 3.7. If X is a CW complex, then $[X, K(\pi, n)]_* = H^n(X; \pi)$. This implies that $K(\pi, n)$ is unique up to homotopy, i.e. $K(\pi, n)$ is canonical in the homotopy category. When $n = 1$ we get that

$$[X, K(\pi, n)]_* = \mathrm{Hom}(\pi_1(X), \pi)$$

Moreover the equality is functorial in X and is induced by $f : X \to K(\pi, n)$ goes to $f_*\iota_n$ where ι_n is the distinguished class in $H^n(K(\pi, n); \pi)$, classified by the identity map $K(\pi, n) \to K(\pi, n)$. Removing the basepoints divides by the conjugation action of π on $\mathrm{Hom}(\pi_1(X), \pi)$.

$$[X, K(\pi, n)] = \prod_{\text{connected components}} \mathrm{Hom}(\pi_1(X_c), \pi)/\operatorname{Ad}(\pi)$$

Corollary 3.8. $\mathrm{Pic}_{\mathbb{R}}(X) \cong H^1(X; \mathbb{Z}/2)$ and $\mathrm{Pic}_{\mathbb{C}}(X) \cong H^2(X; \mathbb{Z})$ given by $L \mapsto w_1(L)$ and $L \mapsto c_1(L)$ respectively.

Beware that there is a choice of sign in the complex case. Note that for us, the distinguished class $\iota_2 \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$ is the Chern class of the dual $\mathcal{O}(1)$ of the tautological line bundle $\mathcal{O}(-1)$.

Remark 3.9. ΣG to BG reduces to $S^2 = \Sigma S^1$ to $\mathbb{CP}^\infty$ but we have to be careful about the orientation, i.e. the choice of $\theta \mapsto e^{i\theta}$ in the S^1 which is getting suspended actually corresponds to $\mathcal{O}(-1)$. This is because we get z which has a pole at infinity, whereas $O(1)$ has a zero infinity.

Proof. Proof that homotopy classes of maps $[X, \mathbb{CP}^\infty]_* = H^2(X; \mathbb{Z})$. The proof is by induction on skeleta. Let X_n be the n-skeleton of X. Write $X_2 = \vee S^2$ because $\mathbb{CP}^\infty$ is simply connected and so X_1 can be contracted. Cellular approximation says that we can just look at maps to $\mathbb{CP}^1$. We get one integer for each 2-cell. This defines a class in $H^2(X_2; \mathbb{Z})$. If that class is zero, we can contract that map from X_2. How to go further?

Inductive step from X_n to X_{n+1}. Suppose first $n \geq 3$. Here $H^2(X_{n+1}, \mathbb{Z}) \to H^2(X_n; \mathbb{Z})$ is an isomorphism. Need to show that maps to $\mathbb{CP}^\infty$ extend uniquely. But this is true because the map $S^n \to \mathbb{CP}^\infty$ is nullhomotopic for $n \geq 3$ and so the map will extend across all disks that we are attaching to get X_{n+1}.

The question is could there be 2-extensions? No. Can deform the guarenteed homotopy to keep it fixed on S^n. This is a topological argument. You have to squint just right.

Now consider $n = 3$. Have a map $H^2(X_3;\mathbb{Z}) \to H^2(X_2;\mathbb{Z})$ which we assume is surjective, i.e. for every $\omega_2 \in H^2(X_2;\mathbb{Z})$ there is a $\omega_3 \in H^2(X_3;\mathbb{Z})$ which restricts to ω_2. We know that for ϕ the attaching map of a 3 disk, we must have $\phi^*\omega_2 = 0 \in H^2(S^2;\mathbb{Z})$. Else, a differential out of ω_2 would block the extension. Induction shows that a map out of ∂D^3 was contractible in $\mathbb{CP}^\infty$ and so argue as before. $\square$

3.3 Additivity

We will give two arguments for the facts

$$w_1(L \otimes L') = w_1(L) + w_1(L')$$
$$c_1(L \otimes L') = c_1(L) + c_1(L')$$

Complex case. Have two maps $\phi, \phi' : X \to \mathbb{CP}^\infty$ classifying L and L' and we want to understand the tensor product bundle. What is the tensor product? It is the pullback of $\mathcal{O}(1,1) = \mathcal{O}(1) \boxtimes \mathcal{O}(1)$ on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ along the map $\phi \times \phi' : X \to \mathbb{CP}^\infty \times \mathbb{CP}^\infty$. The map $t : \mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ classifying $\mathcal{O}(1,1)$. We need to show that $t^*c_1 = c$ where c is the first Chern class of $\mathcal{O}(1,1)$. But then we compute

$$c_1(\mathcal{O}(1,1)) \in H^2(\mathbb{CP}^\infty \times \mathbb{CP}^\infty;\mathbb{Z}) \cong H^2(\mathbb{CP}^\infty;\mathbb{Z}) \oplus H^2(\mathbb{CP}^\infty;\mathbb{Z})$$

agrees with $c_1 \oplus c_1$ because restrictions are $\mathcal{O}(1)$ on each factor. Thus

$$t \circ (\phi \times \phi')^* c_1 = (\phi \times \phi')^*(c_1 \oplus c_1)$$

$\square$

Remark 3.10. Behind the theorem we see that $\mathbb{CP}^\infty$ is up to homotopy a topological abelian group, and the tensor product of line bundles is given by the group structure, i.e. $\mathbb{CP}^\infty \cong \mathrm{Sym}^\infty \mathbb{CP}^1$. For generalized cohomology theories $c_1(L \otimes L')$ defines a 1 parameter formal group, and this quantity $c_1(L \otimes L')$ very nearly determines the cohomology theory. This is due to Quillen and is now christened chromatic homotopy theory.

Definition 3.11. Let V over X be a complex vector bundle of rank r. Form $\mathbb{P}V$ the projectivization of V which is a fiber bundle π over X with fiber $\mathbb{CP}^{r-1}$. Over $\mathbb{P}V$ there is a hyperplane bundle $\mathcal{O}(1)$ which is the dual of the tautological line bundle. Let $\omega = c_1(\mathcal{O}(1))$. Observe that $1, \ldots, \omega^{r-1}$ form a basis for $H^*(\mathbb{P}V_*, \mathbb{Z})$ over each fiber. By the Leray Hirsch theorem

$$H^*(\mathbb{P}V;\mathbb{Z}) \cong \bigoplus_{i=0}^{r-1} \pi^* H^i(X;\mathbb{Z}) \cup \omega^i$$

This implies that there exist a unique relation

$$\omega^r + c_1(V) \cup \omega^{r-1} + \cdots + c_r(V) = 0$$

with $c_i(V) \in H^{2i}(X;\mathbb{Z})$. These are called the Chern classes of V. The sum

$$c(V) = 1 + c_1(V) + \cdots + c_r(V)$$

is called the total Chern class of V.

Theorem 3.12. 1. For $V = L$ the new definition of c_1 agrees with the old one.

2. For any V, the top Chern class $c_r(V)$ is the Euler class of V as a real bundle, i.e. the Thom class of E restricted to the zero section.

3. The Chern classes detect complex conjugation, i.e. $c_i(\overline{V}) = (-1)^i c_i(V)$.

Proof. To check (1) note that $\mathbb{P}V = X$ and the equation is $\omega + c_1(L) = 0$ but $L = \mathcal{O}(-1)$ and so that is ok. Postpone (2) and (3). $\square$

3.4 Characteristic properties

Theorem 3.13. 1. The classes $c_k(V)$ are functorial in V, i.e. for $f : X \to Y$ with V a vector bundle over Y, we have $c_k(f^*V) = f^*c_k(V)$.

2. $c(L) = 1 + c_1(L)$ for line bundles.

Whitney sum For $V = L_1 \oplus L_2$ a direct sum of line bundles, we have $c(V) = c(L_1) \cup c(L_2)$.

3. The above properties completely characterize the Chern classes, i.e. if $c'_k(V)$ is another collection of classes satisfying the above properties, then $c'_k(V) = c_k(V)$ for all k and all V.

Proof. Quick proof of (3). Think about the total space $\mathbb{P}(V \oplus W)$. Inside you have $\mathbb{P}V$ and $\mathbb{P}W$ as disjoint submanifolds. Moreover, observe that $\mathbb{P}(V \oplus W) \setminus \mathbb{P}V$ is a bundle over $\mathbb{P}W$ by projection in homogeneous coordinates. Thus we can compute

$$H^*(\mathbb{P}(V \oplus W)) \cong H^*_{\mathbb{P}W}(\mathbb{P}(V \oplus W))$$

cohomology with supports. Consider now the expression

$$(\sum c_i(V) \cup \omega^{r-i}) \cup (\sum c_j(W) \cup \omega^{s-j})$$

where r, s are the ranks of V, W. The left is 0 in $\mathbb{P}(V)$ and the right is 0 in $\mathbb{P}(W)$, i.e. the left comes from $H^*_{\mathbb{P}W}(\mathbb{P}(V \oplus W))$ and the right comes from $H^*_{\mathbb{P}V}(\mathbb{P}(V \oplus W))$. Thus the cup product is supported nowhere and thus must be zero. But $\sum_m \omega^{r+s-m} \cup c_m(V \oplus W)$ was the only monic equation that could be zero, so we get the desired formula. $\square$

4 Feb 26

4.1 Cohomology of flag varieties

Any universal identity involving chern classes need only be checked on unordered direct sums of line bundles. Given $\pi : V^r \to X$ we get the full flag variety $\mathrm{Fl}(V)$ which is a bundle over X whose points over x are full flags in the fiber V_x. Choose a hermitian metric. Over $\mathrm{Fl}(V)$ we have a filtration of $\pi^* V$ by complete flags with subquotients L_i which gets that $\pi^*(V) \cong \bigoplus L_i$. The metric is what lets us split the filtration, and so in algebraic geometry we cannot necessarily do this. Observe that $\mathrm{Fl}(V)$ is a iterated fiber bundle of projective spaces of decreasing dimension.

Let $x_i \in H^2(\mathrm{Fl}(V); \mathbb{Z})$ be the first Chern class of L_i. An application of Leray Hirsch implies the following

Theorem 4.1. $H^*(\mathrm{Fl}(V); \mathbb{Z}) \cong H^*(X; \mathbb{Z})\langle x_1^{i_1} \dots x_r^{i_r} \rangle$ where $0 \leq i_k \leq r - k$ as a vector space. As an algebra

$$H^*(\mathrm{Fl}(V); \mathbb{Z}) \cong H^*(X; \mathbb{Z})[x_1, \dots, x_r] / \langle \prod (1 + x_i) = c(V) \rangle$$

There is an action of S^r on $H^*(\mathrm{Fl}(V); \mathbb{Z})$ by permuting the x_i and $\pi^* H^*(X; \mathbb{Z}) \hookrightarrow H^*(\mathrm{Fl}(V); \mathbb{Z})$ is the inclusion of the S^r-invariants.

Having a metric, $\mathrm{Fl}(\mathbb{C}^r) \cong U(r)/U(1)^r$ where S^n acts on the right by permuting the columns.

Note that if $X = \mathrm{pt}$, then $V \cong \mathbb{C}^r$ is just a fixed vector space and

$$\mathrm{Fl}(V) = \mathrm{Fl}_r \,.$$

The relation becomes

$$\prod_{i=1}^{r} (1 + x_i) = c(V) = 1$$

since all higher Chern classes of a bundle on a point vanish. Equating homogeneous parts gives

$$e_1(x) = e_2(x) = \cdots = e_r(x) = 0,$$

so the cohomology ring of the full flag variety is the coinvariant algebra of the symmetric group S_r:

$$H^*(\mathrm{Fl}_r; \mathbb{Z}) \cong \mathbb{Z}[x_1, \dots, x_r]/(e_1, \dots, e_r) = \mathbb{Z}[x_1, \dots, x_r] \big/ \langle \mathbb{Z}[x]_+^{S_r} \rangle,$$

For the relative flag bundle, the fiber part has cohomology given by the coinvariants, the base part by the invariants, and the total cohomology is the tensor product of the two.

4.2 Properties of the c_i

$c_k(V^\vee) = (-1)^k c_k(V)$ because we have already seen that the equality holds for c_1 and the splitting principle implies that it holds for all c_k.

The splitting principle also implies that the top Chern class $c_r(V)$ is the Euler class of V as a real bundle. It is true for line bundles and then the Euler class of $V \oplus W$ is the cup product of the Euler classes of V and W.

Proposition 4.2. The Euler class of $V \oplus W$ is the cup product of the Euler classes of V and W.

Proof. We show that it is true for the Thom classes, and recall that the Euler class is the restriction of the Thom class to the zero section. The Thom class is the unique class in $H^{\dim_{\mathbb{R}} V}(D(V), \partial D(V); \mathbb{Z})$ which restricts to the preferred generator in each fiber. V, W being complex have preferred orientation. Clearly $\Theta_v \cup \Theta_w$ satisfies the right property. $\square$

Aside, a priori we have $\Theta_v \cup \Theta_w$ is a class in $H^*(D(V) \times D(W), D(V) \times \partial D(W) \cup \partial D(V) \times D(W); \mathbb{Z})$ but this is isomorphic to $H^*(D(V \oplus W), \partial D(V \oplus W); \mathbb{Z})$. The Euler class can be explicitly represented using distrubtions and differential forms.

Remark 4.3 (Geometric argument for euler product formula)**.** If X is a compact oriented manifold, then $PD(e(V))$ is the zero locus of a generic section of V, which can be made smooth by transversality. Then same for W and $V \oplus W$. The zero loci coincide after pertubation because cup product corresponds to intersection.

Remark 4.4. For a generic transverse section s,

$$Z(s) \subset X$$

is a smooth codimension-k submanifold and

$$PD(e(V)) = [Z(s)] \in H_{n-k}(X).$$

Cup product corresponds to transverse intersection:

$$PD(e(V) \cup e(W)) = [Z(s_V)] \cap [Z(s_W)].$$

Now take $V = N_{X/Y}$. Suppose $i : X \hookrightarrow Y$ is an oriented embedding of codimension k, with normal bundle $N := N_{X/Y}$. There is a fundamental identity:

$$i^* i_*(\alpha) = \alpha \cup e(N) \quad \text{for } \alpha \in H^*(X).$$

This is the self-intersection formula.

The geometric meaning is as follows: i_* corresponds to pushing forward cycles from X into Y, and pulling back i^* amounts to intersecting again with X. The correction factor is the Euler class of the normal bundle. Thus

$$X \cdot X = e(N_{X/Y})$$

inside $H^*(X)$. So cupping with the Euler class of the normal bundle is precisely the operation of intersecting with X inside Y.

Stabliity of chern classes. Observe that $c_*(V \oplus \mathbb{C}) = c_*(V)$ because the extra factor is 1. In general vector bundles over X form a symmetric monoidal category under direct sum and tensor product. Iso classes form a commutative semiring, and we can take an additive group completion to a commutative ring $K^0(X)$, if X is compact and Hausdorff.

The objects in $K^0(X)$ are formal differences of stable isomorphism classes of finite rank vector bundles, where V and W are stably isomorphic if there exists some E such that $V \oplus E \cong W \oplus E$. This can be simplified since every E can be complemented to a trivial bundle, so V and W are stably isomorphic if there exists some n such that $V \oplus \mathbb{C}^n \cong W \oplus \mathbb{C}^n$. Then the Chern classes refine to a map

$$c_* : K^0(X) \to \prod H^{2k}(X; \mathbb{Z})$$

by defining $c_*(V - W) = c_*(V)/c_*(W)$. This is an exponential like map, turning sums into products.

Proposition 4.5. c_* is well defined and turns sums into products. Moreover, if X is compact, the a priori infinite series terminates.

Example 4.6. $X = \mathbb{CP}^\infty$ and consider the class $[-\mathcal{O}(1)] \in K^0(\mathbb{CP}^\infty)$. Then $c_*(-\mathcal{O}(1)) = 1/c_*(\mathcal{O}(1))$ which is $1/(1+c) = 1 - c + c^2 - c^3 + \dots$.

c_* of a tensor product. Let x_i, y_j be the Chern roots of V, W respectively. Then the Chern roots of $V \otimes W$ are $x_i + y_j$. The proof is by the splitting principle.

4.3 Aside on algebraic geometry

In algebraic geometry, we study coherent sheaves. A sheaf S is coherent if locally, there is a presentation of S as the cokernel of a map of finite rank vector bundles.

It is a theorem that on a complex projective manifold, every coherent sheaf has a **finite** resolution by finite rank algebraic vector bundles. In fact, you can stop at the dimension of the manifold.

This is a combination of two theorems. Hilbert syzygy theorem is about doing this locally. The second theorem is about doing this globally, there are enough vector bundles by projectivity.

Use this to functorially define the Chern classes of a coherent sheaf S by taking a resolution by vector bundles and then taking the alternating product of the Chern classes of the vector bundles in the resolution. The result is independent of the choice of resolution. This feeds into the Hirzebruch Riemann Roch theorem.

4.4 Equivariant cohomology and equivariant Euler classes

X space, G compact Lie group acting on X. Can form the Borel construction $X_G = EG \times_G X$ which is a fiber bundle over BG with fiber X. Put $H^*_G(X) = H^*(X_G)$.

Extreme special cases. If G acts trivially on X, then $X_G = BG \times X$ and if BG has no torsion in cohomology, then $H^*_G(X) = H^*(BG) \otimes H^*(X)$.

If G acts freely on X, then $X_G = X/G$ and so $H^*_G(X) = H^*(X/G)$.

Application. If $\pi : V \to X$ is a vector bundle of rank r, it carries a natural $U(1)$ action on the fibers.

Proposition 4.7. The equivariant Euler class $e_{U(1)}(V) \in H^*_{U(1)}(X)$ is

$$\sum_{i=0}^{r} c_i(V) \cup \omega^{r-i}$$

where ω is the generator of $H^*_{U(1)}(\mathrm{pt}) = H^*(\mathbb{CP}^\infty)$.

Proof. V becomes a vector bundle over $X \times \mathbb{CP}^\infty$ by tensoring the pullback with $\mathcal{O}(1)$. The equivariant Euler class is simply the Euler class of this new bundle, a class in $H^*(X \times \mathbb{CP}^\infty)$. Now use the formula for the Chern roots of a tensor product. $\square$

4.5 Hirzebruch Riemann Roch

Note that this generalizes to the Atiyah-Singer index theorem. Let X projective algebraic complex manifold and S a coherent sheaf on X. Denote by $\chi(X, S) = \sum (-1)^i \dim H^i(X, S)$ the Euler characteristic of S. Let t_i denote the Chern roots of the tangent bundle TX, and let s_i be the Choern roots of S. Define the Chern character of S to be

$$\mathrm{ch}(S) = \sum e^{s_i} = \sum_{k=0}^{\infty} \frac{1}{k!} \sum s_i^k$$

in rational cohomology. Define the Todd class of X to be

$$\mathrm{td}(X) = \prod \frac{t_i}{1 - e^{-t_i}} = \prod \frac{t_i}{1 - (1 - t_i + t_i^2/2! - t_i^3/3! + \dots)} = \prod \frac{t_i}{t_i - t_i^2/2! + t_i^3/3! - \dots}$$

Then

$$\chi(X, S) = \int_X \mathrm{ch}(S) \cup \mathrm{td}(X)$$

5 March 3

Today we will do integration along fibers of a bundle. Application will be segre calsses, transgressions of classes, cohomology of $BU(n)$ and $U(n)$, cohomology of $\mathrm{Gr}(k, n)$, and Bott periodicity.

5.1 Integration along the fiber

This is also known as a transfer map, anytime the map goes the wrong way. Let M be compact oriented manifold, then we have a map called capping with the fundamental class $[M]$

$$\int_M : H^*(M) \to H^{*-n}(\mathrm{pt})$$

where n is the dimension of M and this goes the wrong way for cohomology because it goes the same way as the map $M \to *$.

We can do this for any fiber bundle $\pi : E \to B$ with compact oriented fibers of dimension d and the fibers are oriented in a locally constant way.

Theorem 5.1. There is a map $\pi_* : H^*(E) \to H^{*-d}(B)$ where d is the dimension of the fibers, which is functorial in pullback, i.e. given a pullback square

$$\begin{tikzcd} E' \arrow[r] \arrow[d] & E \arrow[d, "\pi"] \\ B' \arrow[r] & B \end{tikzcd}$$

we have $\pi'_* f^* = f^* \pi_*$, and satisfies the projection formula

$$\pi_*(\pi^* \alpha \cup \beta) = \alpha \cup \pi_* \beta$$

Additional functoriality comes in layers of fiber bundles. If $F' \to E' \to B'$ is a fiber bundle and $F \to E \to B$ is a fiber bundle and we have a map of fiber bundle, then we have $(\pi \circ \pi')_* = \pi_* \circ \pi'_*$.

When B is a manifold, this plays well with Poincare duality. Before we define the map, first we give a partial generalization when B is also a manifold. Note that the transfer map could be defined using Poincare duality on both sides and the pushforward of homology classes. This needs not the fiber bundle structure. Let $f : E \to B$ be a proper map of oriented connected manifolds. Note the fibers need not be manifolds. Then there is a functorial $f_* : H^*(E) \to H^{*-d}(B)$ where $d = \dim E - \dim B$ which extends the previous construction when f is a fiber bundle.

Proof. The map is simply the composition

$$\begin{aligned} H^*(E) &\xrightarrow{PD} H_{\dim E - *}(E) \\ &\xrightarrow{f_*} H_{\dim E - *}(B) \\ &\xrightarrow{PD^{-1}} H^{*-d}(B) \end{aligned}$$

where the middle map is the pushforward of homology classes. This relies on the base to be a manifold. Remark that this fails if B is not orietned, then it can be salvaged by twisted coefficients in the orientation sheaf provided f is relatively oriented, meaning $f^*(or_B) \cong or_E$. We can give a second construction by following the map as an embedding followed by projection. If B is finite dimensional, then we can factor f as an proper embedding $E \hookrightarrow B \times \mathbb{R}^N$ followed by projection to B for some large N. Just use Whitney embedding theorem to get the embedding and then use the projection. Then just have to define the pushforward for proper embedding of manifolds. Recall the tubular neighborhood theorem. If $i : X \to Y$ is a proper embedding then there exists $X \hookrightarrow N_{X/Y}$ the normal bundle of X in Y and an open embedding $N_{X/Y} \hookrightarrow Y$ which is the identity on X. The oriented assumption implies that this is oriented. Then Thom implies that we have a Thom isomorphism $H^*(X) \cong H^{*+\dim N_{X/Y}}(D(N_{X/Y}), S(N_{X/Y}))$. Handle the projection using Kunneth formula. $\square$

Remark that the transfer map can operate on the level of differential forms if we have a submersion and the fibers don't pose singularities.

Remark 5.2. Anytime Leray Hirsch holds, then $H^*(E) \cong H^*(B) \otimes H^*(F)$ and the map π_* just integrates the fiber direction.For example, if $u = \sum_i \pi^*(\alpha_i)\, c_i$ with $c_i|_F$ a basis of $H^*(F)$, then

$$\pi_*(u) = \sum_i \alpha_i \int_F c_i|_F$$

In this case, the copy of $H^*(B)$ sits inside $H^*(E)$ as the part with fiber degree 0. Fiber integration projects onto that part by pairing with the top fiber class.

For the flag bundle, this becomes:

$$H^*(\mathrm{Fl}(V)) \cong \bigoplus_I \pi^* H^*(X) \cdot m_I(x_1, \ldots, x_r),$$

and π_* extracts the coefficient of the unique monomial m_I representing the top class of the flag fiber. For $\mathrm{GL}(n)$, the flag fiber is Fl_n and the top class is $x_1^{n-1} x_2^{n-2} \cdots x_{n-1}$. This can also be expressed symbolically as the algebraic operation of taking a polynomial $f \in H^*(X)[x_1, \ldots, x_n]$ and applying

$$\pi_*(f) = \text{coefficient of } x_1^{n-1} x_2^{n-2} \cdots x_{n-1} \text{ in } f = \Delta^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) f(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$$

where $\Delta = \prod_{i<j}(x_i - x_j)$ is the Vandermonde determinant. Note that the sum is alternating and every alternating polynomial is divisible by Δ, whose quotient is a symmetric polynomial (and hence a pullback from the base).

5.2 Application: segre class of a vector bundle

Let $\pi : V \to X$ be a vector bundle and $\mathbb{P}\pi : \mathbb{P}V \to X$ be the projectivization of V. We have the hyperplane bundle $\mathcal{O}(1)$ on $\mathbb{P}V$ and its first Chern class ω.

Definition 5.3. The Segre class of V is defined to be

$$s(V) = \mathbb{P}\pi_* \left(\frac{1}{1-\omega} \right) = \mathbb{P}\pi_*(1 + \omega + \omega^2 + \dots)$$

Theorem 5.4. $s(V) = c(V)^{-1}$, i.e. the Segre class is the multiplicative inverse of the Chern class.

Remark 5.5. In algebraic geometry, can define s of a bundle of projective varieties (cone). In complex AG, things are real codimension 2, thus we get fundamental class and can use this to integrate down.

Proof. **The transfer map is is always a module map over the cohomology of the base**, i.e. $f_*(f^*\alpha \cup \beta) = \alpha \cup f_*\beta$. This is also called the projection formula.

Given this, the theorem is equivalent to the identity

$$\mathbb{P}\pi_*(\frac{1}{1-\omega} \cdot \mathbb{P}\pi^* c(V)) = 1$$

But $1 - \omega$ is the total Chern class of the tautological line bundle $\mathcal{O}(-1)$. Tautological and complement $= \mathbb{P}\pi^* V$. Thus the term $\frac{1}{1-\omega} \cdot \mathbb{P}\pi^* c(V) = c_*(V/\mathcal{O}(-1))$ so it is equal to $1 + c_1 + \dots + c_{r-1}$ where r is the rank of V. The pushforward of this is a constant. Now check it for $\mathbb{P}^{r-1}$ $\square$

5.3 Chern classes of $T\mathbb{P}^n$

What is $T\mathbb{P}^n$? At a point $L \in \mathbb{C}^{n+1}$, the tangent space is the infinitesimal deformations of L inside $\mathbb{C}^{n+1}$. If you draw the picture of a line close to another line, you see that it is the graph of a function from L to $\mathbb{C}^{n+1}/L$. Thus the tangent space is $\mathrm{Hom}(L, \mathbb{C}^{n+1}/L)$. We compute $T\mathbb{P}^n = (\mathbb{C}^{n+1}/L) \otimes L^\vee$ and an exact sequence

$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus n+1} \to T\mathbb{P}^n \to 0$$

where the first map is the diagonal embedding. Now take Chern classes.

$$\begin{aligned} c(T\mathbb{P}^n) &= c(\mathcal{O}(1)^{\oplus n+1})/c(\mathcal{O}) \\ &= (1+c)^{n+1} \end{aligned}$$

In particular $c_n(T\mathbb{P}^n) = \binom{n+1}{n} c^n = (n+1)c^n$ is the Euler class and we can check that

$$\int_{\mathbb{P}^n} c_n(T\mathbb{P}^n) = n + 1 = \chi(\mathbb{P}^n)$$

5.4 Cohomology of $\mathrm{Gr}(k, n)$

We have bundles

$$0 \to T \to \underline{\mathbb{C}}^n \to Q \to 0$$

where T is the tautological bundle of rank k and Q is the quotient bundle of rank $n - k$. We have classes $c_i(T)$ and $c_j(Q)$ and the identity

$$c(T) \cup c(Q) = 1$$

The first observation is that these generators are redundant. In fact, $c_i(Q)$ can be expressed in terms of $c_j(T)$ and vice versa, e.g. $c_1(Q) = -c_1(T)$. One might worry that if one throws away half of the generators, then one might lose the relations. The second observation is that this is no issue, because relations are recovered from the condition that $c(T) \cup c(Q) = 1$. The third observation is that there are no other relations.

Proof. There is a fiber bundle $\mathrm{Fl}(k) \times \mathrm{Fl}(n-k) \hookrightarrow \mathrm{Fl}(n) \to \mathrm{Gr}(k, n)$ so

$$H^*(\mathrm{Gr}(k, n)) = [\mathbb{Z}[x_i]/(e_i = 0)]^{S^k \times S^{n-k}}$$

The S_k invariants correspond to $c_i(T)$ and the S_{n-k} invariants correspond to $c_j(Q)$. $\square$

More generally for $\pi : V \to X$ and $\mathrm{Gr}\,\pi : \mathrm{Gr}(k, V) \to X$ the Grassmannian bundle, we have $H^*(\mathrm{Gr}(k, V)) = H^*(X)\langle c_i(T), c_j(Q)\rangle / \langle c(T) \cup c(Q) = c(V)\rangle$.

5.5 Cohomology of $BU(n)$ and $U(n)$

Write

$$BU(n) = \mathrm{Gr}(n, \infty)$$

from which it becomes clear that

$$H^*(BU(n); \mathbb{Z}) \cong \mathbb{Z}[c_1, \ldots, c_n]$$

where c_i are the universal chern classes, in the sense that any vector bundle V over X is classified by a map $f : X \to BU(n)$ and $c_i(V) = f^* c_i$.

Now we can do transgression to get the cohomology of $U(n)$. Consider transgression on a mapping space with source M and target X. Produce cohomology classes in $H^*(\mathrm{Map}(M, X))$ so called tautological classes. There is an evaluation map $\mathrm{Map}(M, X) \times M \to X$. Given $\alpha \in H^*(X)$, we can pull it back to $\mathrm{Map}(M, X) \times M$ and then use Kunneth to get components in $H^*(M)$.

Fun example where $X = BU(n)$ and $M = \Sigma$ is a closed surface. Get tautological classes in $H^*(\mathrm{Map}(\Sigma, BU(n)))$ from c_i and a basis of $H^*(\Sigma)$.

Theorem 5.6 (Atiyah-Bott, Proceedings of Royal Society, 1980s, "Yang Mills")**.** In this case, the cohomology of $\mathrm{Map}(\Sigma, BU(n))$ is freely generated by these classes as an algebra.

For us, consider the case $M = S^1, *$. Consider the maps

$$\begin{aligned}(S^1, *) \times \Omega BU(n) &\to BU(n)\\ S^1 \wedge U(n) &\to BU(n)\end{aligned}$$

where keep in mind $U(n) = \Omega BU(n)$ homotopy equivaelnt. We have the fiber bundle $U(n-1) \to U(n) \to S^{2n-1}$ where the last map is claled q. Then

Theorem 5.7.

$$\int_{S^1} ev^* c_i = q^* vol_{S^{2n-1}}$$

Corollary 5.8. The classes in $H^{2k-1}(U(n))$ pulled back from $H^{2k-1}(S^{2k-1})$are defined on any $U(n)$ for $n \geq k$. This implies the Leray assumption and we can compute $H^*(U(n))$ as the exterior algebra on the transgressions of the c_i.

6 March 10

6.1 Symmetric functions and grassmannians

There is a duality isomorphism

$$\begin{gathered}\mathrm{Gr}(k, n) \cong \mathrm{Gr}(n-k, n)\\ (T \to \mathbb{C}^n \to Q) \mapsto (\overline{T}^\perp \to \overline{C^n} \to \overline{Q}^\perp)\end{gathered}$$

note that $T \mapsto T^\perp$ is complex antilinear, so we need to conjugate to get a complex linear map. The chern roots change sign.

Theorem 6.1. D is compattible with Schubert stratification. On labels, one exchanges 0s and 1s and reverses the sequence. On partitions, one takes the dual. If λ fits in the $k \times (n-k)$ box, then λ^D will fit in the $(n-k) \times k$ box.

Why do Schubert cells map to cells? We need to show that if $\dim(T \cap F_q)$ jumps at $q = i_1, i_2, \ldots,$ then $\dim(T^\perp \cap F_q^D)$ jumps at $n - i_1, n - i_2, \ldots$. This is a linear algebra fact.

This duality is also witnessed by an algebraic duality in the ring of symmetric polynomials. Let R be the ring of symmetric polynomials in infinite variables.

Theorem 6.2. The map $s_\lambda \in R \mapsto PD(\Sigma_\lambda) \in H^{2|\lambda|}(\mathrm{Gr}(k,n))$ is a surjective ring hommorphism. The homogeneous variables are either the Chern roots of Q or the negative Chern roots of T. They are interchanged by the duality isomorphism and it interchanges the partition labels. In particular

$$PD(\Sigma_\lambda) \cdot PD(\Sigma_\mu) = \sum_\nu c_{\lambda\mu}^\nu PD(\Sigma_\nu)$$

are the Littlewood-Richardson coefficients, which coincide with the structure constants of the Schur functions. Using the negatiev Chern roots of T we can also write

$$\begin{aligned} c_i(T) &= (-1)^i e_i(x_1, \ldots, x_k) = (-1)^i S_{(1^i)} \\ c_j(Q) &= s_j(x_1, \ldots, x_k) = S_{(j)} \end{aligned}$$

where e_i is the i-th elementary symmetric function and s_j is the j-th complete symmetric function.

A proof of this theorem follows from the Pieri formula.

Theorem 6.3. The Pieri formula states that

$$c(Q) \cap [\Sigma_\lambda] = \sum_\nu [\Sigma_\nu]$$

where the sum is over ν obtained from λ by adding a horizontal strip, i.e. adding boxes to λ with no two in the same column.

Remark 6.4. Schur Weyl duality considers the two sided action of GL_k and S_n on $(\mathbb{C}^k)^{\otimes n}$.

Constantin doesn't want to do the proofs so we will just explain the logic. The geometric proof of Pieri. Look at $\mathrm{Gr}(k,n)$ and understand the multiplication of $c_*(Q)$ on homogeneous polynomials. Pass to the flag bundle of Q over $\mathrm{Gr}(k,n)$. Every Schubert variety Σ_λ has a rational section, regular over the Schubert cell. This is because Q can be trivialized over the Schubert cell, the basis coming from restricting the total coordinates of $C^n \to Q$ complimenting the pivot coordinates. Use the identity $p_*(p^* c(Q) \cap \sigma_\lambda) = c(Q) \cap p_* \sigma_\lambda$ where upstairs you have a formula.

Remark 6.5. Similar idea can be used to prove that $[\Sigma_\lambda]$ is the Schur function s_λ in the Chern roots of Q. Use $\mathrm{Fl}(k) \to \mathrm{Fl}(T) \to \mathrm{Gr}(k,n)$ with Chern roots ρ_i of T for $i = 1, \ldots, k$. The fact is that $\mathrm{Fl}(T)$ has a stratification lifting the one in $\mathrm{Gr}(k)$. There are $k!$ strata over each Σ_λ, namely the strata of $\mathrm{Fl}(k)$. $\mathrm{Fl}(k)$ is an iterated fiber bundle of projective spaces. The fact is $PD(\Sigma_\lambda) = p_* PD(\rho_1^{\lambda_1+k-1} \cup \cdots \cup \rho_k^{\lambda_k})$.

The wrong way map in cohomology here becomes the correct map in homology, so we get the zero locus of maps generically bijective to Σ_λ.

Recall that we have the Jacobi-Trudi formula

$$\sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{k} s_{\lambda_i+n-i} = \frac{A}{B}$$

where A is the determinant of the matrix with (i,j)-entry s_{λ_i+j-i} and B is A with $\lambda_i = 0$. Then the fact above is about the application of the Jacobi-Trudi formula to monomials.

Repeating the argument in K thoery gives the character formula for representations of $\mathrm{GL}(n)$ as Schur functions. We get a bunch of theorems

Theorem 6.6.

1. $H^0(\mathrm{Fl}(k), L_\lambda) = S_\lambda$ is the highest weight representation of $\mathrm{GL}(k)$ with highest weight λ.
2. (Bott improvement) $H^{>0} = 0$ if λ is dominant.
3. $\sum(-1)^i H^i$ conincudes with the pushforward $p_!$ in K theory
4. $p_!$ in K theory obeys the same formula.

6.2 Bott periodicity theorem

We didn't give the definition of homotopy groups yet but we can give a weak version of the statement.

Theorem 6.7. $\pi_k(U) = 0$ for k even and $\pi_k(U) = \mathbb{Z}$ for k odd, where $U = \lim U(n)$ is the infinite unitary group.

Let's state the better version.

Theorem 6.8. There is a homotopy equivalence

$$\beta : BU \to \Omega^0 U$$

where $\Omega^0 U$ is the connected component of the based loop space of U containing the constant loop. In particular, $\Omega U \simeq \mathbb{Z} \times BU$.

Remark 6.9. Grouplike monoid structure of ΩU (any space honestly) means that $\Omega U \simeq \mathbb{Z} \times \Omega^0 U$ where $\mathbb{Z}$ is the fundamental group of U. To see this, note that as a space $U(n)$ is homeomorphic to $U(1) \times SU(n)$ the map to the first component being the determinant, and the map is split by $u \in \mathbb{R} \mapsto \mathrm{diag}(u, 1, \ldots, 1)$.

We can state the Bott map. Realize BU as limit of Grassmannian. Note that $\Omega SU(n)$ is maps $I \to SU(n)$ sending the endpoints to the identity. We can consider $P_{I,-I}(SU(2n))$ the space of paths in $SU(2n)$ sending the endpoints to $\pm I$. Note that $P_{I,-I}(SU(2n))$ is homotopy equivalent to $\Omega SU(2n)$ because we can simply choose any path from I to $-I$ and concatenate.

This receives a map from $\lim \mathrm{Gr}(n, 2n)$. The map will go to minimal geodesics in $SU(2n)$. We will send $T \in \mathrm{Gr}(n, 2n)$ as follows. Write $\mathbb{C}^{2n} = T \oplus T^{\perp}$ and define a path in $SU(2n)$ by sending $t \in I$ to $e^{\pi i t} \operatorname{Proj}_T + e^{-\pi i t} \operatorname{Proj}_{T^{\perp}}$.

Theorem 6.10. The image is a union of cells in a cell decomposition of a cellular approximation of $P_{I,-I}(SU(2n))$. The other cells has dimension $2n + 2$. In particular we get all the lower cells.

Sources of the cell decomposition.

1. Morse theory a la Bott.

2. Algebra. Coarsening of the Bruhat decomposition of ΩSU.

Part of the story is that there is an algebraic version of ΩU. We can consider the Laurent polynomial loop group and its flag variety. $L_{poly} \mathrm{GL}_n(\mathbb{C})$ is the subgroup of invertibles in $\mathrm{Mat}_n(\mathbb{C}[z, z^{-1}])$. Note that $U(n) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$ is a homotopy equivalence by the polar decomposition because M invertible implies that $M = up$ and P the positive definite part is contractible.

Theorem 6.11. $L_{poly} \mathrm{GL}_n(\mathbb{C}) \hookrightarrow L \mathrm{GL}_n(\mathbb{C})$ is a homotopy equivalence. The right hand side is a CW complex because you can filter by the order of zeros and poles.

We can get this form Bott or an analog of Schubert decomposition. We consider Bott's argument. Morse theory. Consider the energy functional on $P_{I,-I}(SU(2n))$ given by $E(\gamma) = \int_0^1 \|\gamma'(t)\|^2 dt$. The critical points are precisely the geodesics from I to $-I$. Note that $L_{sm}GL(n) \subset L_{cts}GL(n)$ is a homotopy equivalence by general nonsense so thinking about derivatives is harmless. The critical points are precisely the image of the map $\lim \mathrm{Gr}(n, 2n) \to P_{I,-I}(SU(2n))$. Gradient flow defines a cell decomposition of $P_{I,-I}(SU(2n))$. The dimension of the cells is computable from the Hessian of E.